

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	17 July 20226
Team ID	
Project Name	<i>Rising Waters: A Machine Learning Approach to Flood Prediction</i>
Maximum Marks	4 Marks

Functional Requirements:

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system shall provide secure user authentication for administrators using username and password. General users can access the prediction interface without login.	High
2	Authorization Levels	The system shall support role-based access. Administrator can manage datasets, retrain models, and deploy updates. Users can enter weather parameters and view flood predictions.	High
3	External Interfaces	The system shall integrate with historical flood datasets (CSV), Python ML libraries (Scikit-learn, XGBoost), Flask framework, and may optionally integrate with weather APIs (e.g., OpenWeatherMap) for real-time data.	High
4	Transaction Processing	The system shall accept rainfall and weather inputs, preprocess the data, load the trained XGBoost model, generate	High

S.No	Requirement Category	Requirement Description	Priority
		predictions, and display flood risk instantly.	
5	Reporting	The system shall generate prediction results, model evaluation metrics (Accuracy, Precision, Recall, F1-Score), confusion matrix, and data visualizations for analysis.	Medium
6	Business Rules	Only valid numerical weather parameters shall be accepted. Predictions shall use the trained XGBoost model. Missing or invalid inputs must be rejected with appropriate validation messages.	High
7	Compliance to Laws or Regulations	The system shall comply with data privacy principles by avoiding the collection of personal user information and using only publicly available environmental datasets for model training.	Medium
8	Other	The system shall provide a simple, responsive Flask web interface that allows users to obtain flood predictions quickly and easily.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

S.N o	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system shall generate flood prediction results with minimal delay after user input.	Prediction response time < 2 seconds
2	Scalability	The application shall support increasing datasets and multiple users without significant performance degradation.	Supports 100+ simultaneous users and future dataset expansion
3	Security & Data Privacy	User inputs shall be validated, administrator credentials protected, and the application secured against common web vulnerabilities.	Input validation, HTTPS support, secure password storage
4	Reliability & Availability	The application shall provide consistent prediction results and remain available during normal operating conditions.	99% application availability
5	Usability & Accessibility	The web application shall provide an intuitive user interface with clearly labeled input fields and prediction results accessible on desktop and mobile browsers.	Easy navigation, responsive design, browser compatibility
6	Maintainability	The system shall be modular,	Modular

S.N o	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
	ty & Portability	well-documented, and portable across Windows, Linux, and cloud platforms. Future ML models can be integrated with minimal code changes.	architecture with reusable Python and Flask components